

FitBuddy

Phase 5: Project Development Phase

Smart Fitness and Wellness Application

1. Development Approach

FitBuddy is developed iteratively: frontend, backend, and database are built and integrated module by module, with the recommendation system added once the core flow works.

2. Frontend Development

- HTML pages for registration, login, profile, dashboard, workouts, nutrition, activity log, and progress
- CSS for a responsive, clean and consistent layout
- JavaScript for form validation, dynamic content, and progress charts

3. Backend Development (Python)

- User authentication and session management
- REST endpoints for profile, goals, workouts, activities, and progress
- BMI and calorie calculation functions
- Recommendation engine processing age, weight, goal, and activity level

4. Database Implementation (MySQL)

Tables from the design phase are created and populated with workout and nutrition data. The backend connects using a MySQL connector with parameterized queries.

```
CREATE TABLE users (user_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100), email VARCHAR(100) UNIQUE, password_hash VARCHAR(255), age INT, height FLOAT, weight FLOAT, activity_level VARCHAR(20));
```

5. Sample Recommendation Logic

```
def calc_bmi(weight, height_m): return weight / (height_m ** 2)
def recommend(goal, bmi, level):
    if goal == 'weight_loss': return 'cardio', 'medium' if level != 'sedentary' else 'low'
    if goal == 'muscle_gain': return 'strength', 'medium'
    return 'mixed', 'low'
```

6. Feature Integration

Module	Status Checkpoint
Authentication	Users can register and log in securely
Profile and Goals	Data saved and editable
Recommendations	Suggestions generated from profile
Activity Logging	Records stored and retrievable
Progress Tracking	Charts reflect stored data

7. Coding Standards

Use meaningful names, comment key logic, follow modular structure, and commit regularly to version control.